

7.12. Attribute

An attribute is cluster data. Each attribute SHALL be listed in a table with [data quality](#) columns: ID, Name, (Data) Type, Constraint, other Quality, Access, Default (value), and Conformance. An attribute SHALL also define its associated semantics and behavior. Attributes reflect queryable/settable state, configuration and capabilities of a device. If no privileges are explicitly defined for an attribute, then [default access privileges](#) take effect. Attribute data MAY also have these other qualities:

Quality	Short	Description
Scene	S	indicates that the data is part of a scene
Persistent	N	indicates that the data value is persistent across restarts
Fixed	F	indicates that the read only value is a value that changes infrequently (see description)
Nullable	X	indicates that the data may have a value of NULL
Atomic	T	indicates that the attribute may only be written to in the context of an atomic write

Fabric-scoped attribute data SHALL be defined as a [fabric-scoped list](#).

7.12.1. Persistence

Persistent data retains its value across a restart.

A restart is:

- a program restart or reboot
- power cycle reboot
- user-initiated reboot
- reboot initiated from a program image upgrade

A factory reset is not such a restart. A factory reset is a deliberate behavior to reset persistent data back to its original state when the product left the factory.

Cluster attributes that represent configuration data SHALL be persistent data unless otherwise specified.

For example: a writable attribute that persistently changes the behavior (or mode) of the cluster.